

AIM: Automated Glass Detection and Water Dispensing System using Arduino

Introduction

The Automated Glass Detection and Water Dispensing System using Arduino is an IoT-based smart automation project that detects the presence of a glass using an ultrasonic sensor and fills it with water upon detection. This project integrates an IR sensor to detect the water level in the glass and a servo motor to control the water flow. By utilizing sensor-based automation, the system ensures efficient and contactless water dispensing, making it suitable for public spaces, offices, and homes.

Key Features

Automated Glass Detection: Uses an ultrasonic sensor (HC-SR04) to detect when a glass is placed in position.

Water Level Monitoring: The IR sensor ensures water fills up to the desired level and prevents overflow.

Servo Motor Control: The servo motor regulates water flow by opening/closing a valve when needed.

Energy Efficient: The system only operates when a glass is detected, conserving energy and water.

User-Friendly and Contactless: No physical buttons are needed, making it hygienic and convenient.

Working Principle

1. Glass Detection (Ultrasonic Sensor):

- The **ultrasonic sensor (HC-SR04)** continuously measures distance.
- If the distance is **less than 20 cm**, the system detects a glass and triggers the servo motor.

2. Water Dispensing (Servo Motor):

- When a glass is detected, the **servo motor rotates**, allowing water to flow.

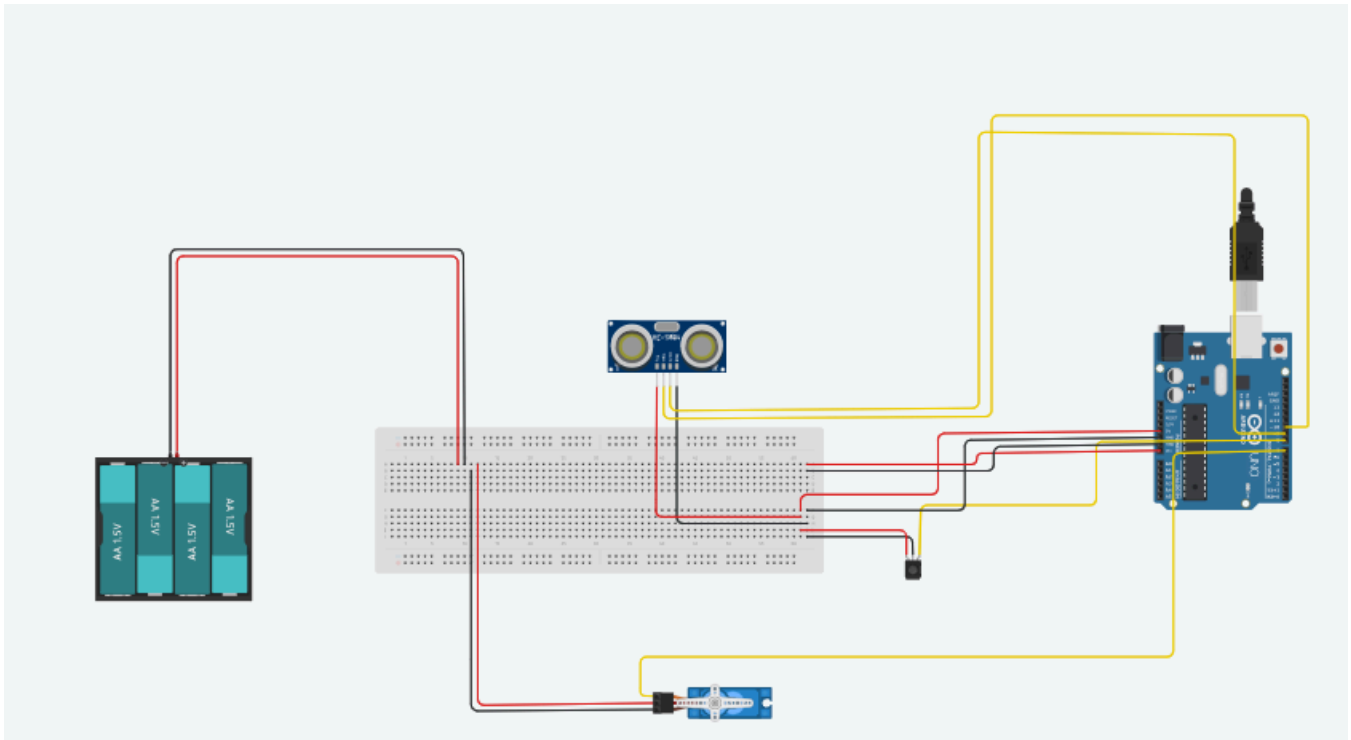
3. Water Level Detection (IR Sensor):

- The **IR sensor** monitors the water level.
- Once the glass is **full**, the IR sensor sends a signal to the Arduino, and the **servo motor closes the valve**, stopping the water flow.

4. Idle Mode:

- If no glass is detected, or after filling is completed, the system resets and waits for the next glass.

Circuit Diagram:



- ◆ **Arduino Uno:** Central microcontroller that processes sensor data and controls the servo motor.
- ◆ **Ultrasonic Sensor (HC-SR04):** Detects the presence of a glass.
- ◆ **IR Sensor:** Detects when the glass is full.
- ◆ **Servo Motor:** Opens/closes the water valve.
- ◆ **Battery Pack:** Powers the circuit.
- ◆ **Jumper Wires & Breadboard:** Used for connections.

Advantages

- ✓ **Touchless & Hygienic:** Reduces contamination, making it ideal for public places.
- ✓ **Water Conservation:** Prevents spillage and overflow using the IR sensor.
- ✓ **Energy Efficient:** Uses minimal power, making it suitable for long-term use.
- ✓ **Easy to Use & Maintain:** Simple plug-and-play operation with minimal setup.

Conclusion

The Automated Glass Detection and Water Dispensing System provides a contactless and efficient way to dispense water, eliminating the need for manual operation. By combining ultrasonic and IR sensors with a servo motor, the system ensures precise filling and prevents water wastage. This project is an excellent demonstration of sensor-based automation and IoT applications, making it ideal for public spaces, offices, restaurants, and homes.